

Hidden Markov Models for Financial Regime Detection: A Rigorous Treatment of the Baum-Welch Algorithm and Inference Procedures

Quantitative Analysis

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Abstract

We present a rigorous mathematical treatment of Hidden Markov Models (HMMs) applied to financial regime detection, specifically for the Nifty 50 index. The document covers model specification, maximum likelihood estimation via the Baum-Welch expectation-maximization algorithm, state classification methodology, and both filtering (causal) and smoothing (non-causal) inference procedures. Special attention is given to numerical stability via log-space computations and the economic interpretation of latent states as Bull, Bear, and Crisis regimes.

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1 Model Specification

1.1 The Hidden Markov Model

Let $\{Y_t\}_{t=1}^T$ be a sequence of observed random variables representing daily log-returns of a financial asset, where $Y_t \in \mathbb{R}$. Let $\{S_t\}_{t=1}^T$ be a sequence of hidden (latent) random variables taking values in the finite state space $\mathcal{S} = \{1, 2, \dots, K\}$ with $K = 3$.

Definition 1.1 (Hidden Markov Model). A Hidden Markov Model is characterized by the parameter vector $\boldsymbol{\lambda} = (\boldsymbol{\pi}, \mathbf{A}, \boldsymbol{\theta})$ where:

1. $\boldsymbol{\pi} = (\pi_1, \dots, \pi_K)$ is the initial state distribution with $\pi_k = P(S_1 = k)$ and $\sum_{k=1}^K \pi_k = 1$.
2. $\mathbf{A} = [a_{ij}] \in [0, 1]^{K \times K}$ is the transition matrix with $a_{ij} = P(S_t = j \mid S_{t-1} = i)$ satisfying $\sum_{j=1}^K a_{ij} = 1$ for all i .
3. $\boldsymbol{\theta} = \{(\mu_k, \sigma_k^2)\}_{k=1}^K$ are the emission parameters.

The joint probability of a state sequence $\mathbf{s}_{1:T} = (s_1, \dots, s_T)$ and observations $\mathbf{y}_{1:T} = (y_1, \dots, y_T)$ factorizes as:

$$p(\mathbf{y}_{1:T}, \mathbf{s}_{1:T} \mid \boldsymbol{\lambda}) = \pi_{s_1} b_{s_1}(y_1) \prod_{t=2}^T a_{s_{t-1}, s_t} b_{s_t}(y_t) \quad (1)$$

where $b_k(y_t) = p(y_t \mid S_t = k)$ is the emission density.

1.2 Gaussian Emissions

For financial returns, we assume Gaussian emissions:

$$b_k(y_t) = \frac{1}{\sqrt{2\pi\sigma_k^2}} \exp\left(-\frac{(y_t - \mu_k)^2}{2\sigma_k^2}\right) \quad (2)$$

Thus $Y_t \mid S_t = k \sim \mathcal{N}(\mu_k, \sigma_k^2)$. The complete parameter set is $\boldsymbol{\lambda} = (\boldsymbol{\pi}, \mathbf{A}, \boldsymbol{\mu}, \boldsymbol{\sigma}^2)$.

2 Parameter Estimation

Given observations $\mathbf{y}_{1:T}$, we seek the maximum likelihood estimate:

$$\hat{\boldsymbol{\lambda}} = \arg \max_{\boldsymbol{\lambda}} \mathcal{L}(\boldsymbol{\lambda}) = \arg \max_{\boldsymbol{\lambda}} \log p(\mathbf{y}_{1:T} \mid \boldsymbol{\lambda}) \quad (3)$$

where the likelihood is obtained by marginalizing over all possible state sequences:

$$p(\mathbf{y}_{1:T} \mid \boldsymbol{\lambda}) = \sum_{\mathbf{s}_{1:T} \in \mathcal{S}^T} p(\mathbf{y}_{1:T}, \mathbf{s}_{1:T} \mid \boldsymbol{\lambda}) \quad (4)$$

Direct computation of (4) is $O(K^T)$. We use the Baum-Welch algorithm (special case of EM).

2.1 The Q-Function

Let $\boldsymbol{\lambda}^{(n)}$ be the parameter estimate at iteration n . The complete-data log-likelihood is:

$$\log p(\mathbf{y}_{1:T}, \mathbf{s}_{1:T} \mid \boldsymbol{\lambda}) = \sum_{k=1}^K \mathbb{K}_{[s_1=k]} \log \pi_k + \sum_{t=2}^T \sum_{i,j} \xi_t(i, j) \log a_{ij} + \sum_{t=1}^T \sum_{k=1}^K \gamma_t(k) \log b_k(y_t) \quad (5)$$

where $\gamma_t(k) = \mathbb{K}_{[s_t=k]}$ and $\xi_t(i, j) = \mathbb{K}_{[s_{t-1}=i, s_t=j]}$.

Since $\mathbf{s}_{1:T}$ is unobserved, we maximize the expected complete-data log-likelihood (Q-function):

$$Q(\boldsymbol{\lambda}, \boldsymbol{\lambda}^{(n)}) = \mathbb{E}_{\mathbf{S} \mid \mathbf{y}_{1:T}, \boldsymbol{\lambda}^{(n)}} [\log p(\mathbf{y}_{1:T}, \mathbf{S} \mid \boldsymbol{\lambda})] \quad (6)$$

2.2 E-Step: Forward-Backward Algorithm

Define the forward and backward variables:

Definition 2.1 (Forward Variables).

$$\alpha_t(k) = p(y_1, \dots, y_t, S_t = k \mid \boldsymbol{\lambda}) \quad (7)$$

satisfying the recursion:

$$\alpha_1(k) = \pi_k b_k(y_1) \quad (8)$$

$$\alpha_t(k) = b_k(y_t) \sum_{i=1}^K \alpha_{t-1}(i) a_{ik}, \quad t = 2, \dots, T \quad (9)$$

Definition 2.2 (Backward Variables).

$$\beta_t(k) = p(y_{t+1}, \dots, y_T \mid S_t = k, \boldsymbol{\lambda}) \quad (10)$$

satisfying the recursion:

$$\beta_T(k) = 1 \quad (11)$$

$$\beta_t(k) = \sum_{j=1}^K a_{kj} b_j(y_{t+1}) \beta_{t+1}(j), \quad t = T-1, \dots, 1 \quad (12)$$

Proposition 2.1 (Smoothed Posteriors). *The posterior probabilities are given by:*

$$\gamma_t(k) = P(S_t = k \mid \mathbf{y}_{1:T}, \boldsymbol{\lambda}) = \frac{\alpha_t(k) \beta_t(k)}{\sum_{j=1}^K \alpha_t(j) \beta_t(j)} \quad (13)$$

$$\xi_t(i, j) = P(S_{t-1} = i, S_t = j \mid \mathbf{y}_{1:T}, \boldsymbol{\lambda}) = \frac{\alpha_{t-1}(i) a_{ij} b_j(y_t) \beta_t(j)}{\sum_{k=1}^K \alpha_{t-1}(k) \beta_t(k)} \quad (14)$$

Proof. By Bayes' rule and the conditional independence structure of the HMM:

$$\begin{aligned} \gamma_t(k) &= \frac{p(\mathbf{y}_{1:T}, S_t = k)}{p(\mathbf{y}_{1:T})} = \frac{p(y_1, \dots, y_t, S_t = k) p(y_{t+1}, \dots, y_T \mid S_t = k)}{p(\mathbf{y}_{1:T})} \\ &= \frac{\alpha_t(k) \beta_t(k)}{\sum_j \alpha_t(j) \beta_t(j)} \end{aligned}$$

The denominator follows from $\sum_k P(S_t = k \mid \mathbf{y}_{1:T}) = 1$. The derivation for $\xi_t(i, j)$ follows similarly by considering the joint probability of consecutive states. $\square$

2.3 Log-Space Computation

To prevent numerical underflow, we compute in log-space. Define $\tilde{\alpha}_t(k) = \log \alpha_t(k)$.

Algorithm 1 Log-Forward Algorithm

- 1: $\tilde{\alpha}_1(k) = \log \pi_k + \log b_k(y_1)$ for all k
 - 2: **for** $t = 2$ to T **do**
 - 3: **for** $j = 1$ to K **do**
 - 4: $\tilde{\alpha}_t(j) = \log b_j(y_t) + \text{logsumexp}_{i=1}^K(\tilde{\alpha}_{t-1}(i) + \log a_{ij})$
 - 5: **end for**
 - 6: **end for**
 - 7: where $\text{logsumexp}(\mathbf{x}) = \log(\sum_i e^{x_i}) = x_{\max} + \log(\sum_i e^{x_i - x_{\max}})$
-

2.4 M-Step: Parameter Updates

Maximizing $Q(\boldsymbol{\lambda}, \boldsymbol{\lambda}^{(n)})$ yields closed-form updates:

Proposition 2.2 (M-Step Updates).

$$\hat{\pi}_k = \gamma_1(k) \quad (15)$$

$$\hat{a}_{ij} = \frac{\sum_{t=2}^T \xi_t(i, j)}{\sum_{t=2}^T \gamma_{t-1}(i)} \quad (16)$$

$$\hat{\mu}_k = \frac{\sum_{t=1}^T \gamma_t(k) y_t}{\sum_{t=1}^T \gamma_t(k)} \quad (17)$$

$$\hat{\sigma}_k^2 = \frac{\sum_{t=1}^T \gamma_t(k) (y_t - \hat{\mu}_k)^2}{\sum_{t=1}^T \gamma_t(k)} \quad (18)$$

Proof. The updates follow from setting partial derivatives of Q to zero subject to constraints $\sum_k \pi_k = 1$ and $\sum_j a_{ij} = 1$, using Lagrange multipliers. For $\hat{\mu}_k$:

$$\frac{\partial Q}{\partial \mu_k} = \sum_{t=1}^T \gamma_t(k) \frac{(y_t - \mu_k)}{\sigma_k^2} = 0 \implies \hat{\mu}_k = \frac{\sum_t \gamma_t(k) y_t}{\sum_t \gamma_t(k)} \quad (19)$$

The variance update follows similarly from $\partial Q / \partial \sigma_k^2 = 0$. □

The algorithm iterates E-step and M-step until convergence: $|\mathcal{L}^{(n+1)} - \mathcal{L}^{(n)}| / |\mathcal{L}^{(n)}| < \epsilon$.

3 State Classification: Economic Ordering

The HMM fitting yields unlabeled states $\{1, 2, 3\}$. We impose economic structure via:

Definition 3.1 (Economic Classification). Let $\hat{\boldsymbol{\mu}} = (\hat{\mu}_1, \hat{\mu}_2, \hat{\mu}_3)$ be the estimated means. Define the permutation $\rho : \{1, 2, 3\} \rightarrow \{1, 2, 3\}$ such that:

$$\hat{\mu}_{\rho(1)} \geq \hat{\mu}_{\rho(2)} \geq \hat{\mu}_{\rho(3)} \quad (20)$$

The economic labels are assigned as:

Rank	Label	Interpretation
1	Bull	$\hat{\mu}_{\text{bull}} = \max_k \hat{\mu}_k > 0$
2	Bear	$\hat{\mu}_{\text{bear}} =_k \hat{\mu}_k$
3	Crisis	$\hat{\mu}_{\text{crisis}} = \min_k \hat{\mu}_k$

All parameters are remapped: $\hat{\mu}_k^{\text{sorted}} = \hat{\mu}_{\rho(k)}$, $\hat{\mathbf{A}}^{\text{sorted}} = \mathbf{P}_\rho \hat{\mathbf{A}} \mathbf{P}_\rho^\top$ where $\mathbf{P}_\rho$ is the permutation matrix.

4 Decoding and Inference

4.1 Viterbi Algorithm: Global Decoding

To find the most likely state sequence $\mathbf{s}^* = \arg \max_{\mathbf{s}} P(\mathbf{s} \mid \mathbf{y})$, we use dynamic programming.

Definition 4.1 (Viterbi Variables). Let $\delta_t(k) = \max_{s_1, \dots, s_{t-1}} P(s_1, \dots, s_t = k, y_1, \dots, y_t)$.

Algorithm 2 Viterbi Algorithm

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1:  $\delta_1(k) = \pi_k b_k(y_1)$ 
2: for  $t = 2$  to  $T$  do
3:   for  $k = 1$  to  $K$  do
4:      $\delta_t(k) = b_k(y_t) \cdot \max_i [\delta_{t-1}(i) \cdot a_{ik}]$ 
5:      $\psi_t(k) = \arg \max_i [\delta_{t-1}(i) \cdot a_{ik}]$ 
6:   end for
7: end for
8:  $s_T^* = \arg \max_k \delta_T(k)$ 
9: for  $t = T - 1$  downto  $1$  do
10:   $s_t^* = \psi_{t+1}(s_{t+1}^*)$ 
11: end for
```

Complexity is $O(TK^2)$ versus $O(K^T)$ for brute force.

4.2 Filtering vs. Smoothing

Definition 4.2 (Filtering). The filtered distribution uses only past and present data:

$$\hat{\alpha}_t(k) = P(S_t = k \mid \mathbf{y}_{1:t}) = \frac{\alpha_t(k)}{\sum_j \alpha_t(j)} \quad (21)$$

This is causal and suitable for real-time applications.

Definition 4.3 (Smoothing). The smoothed distribution uses all data:

$$\gamma_t(k) = P(S_t = k \mid \mathbf{y}_{1:T}) \quad (22)$$

This is non-causal (uses future) and optimal for historical analysis.

5 Prediction

At time T , we observe $\mathbf{y}_{1:T}$ and wish to predict Y_{T+1} .

5.1 Regime Prediction

The one-step-ahead regime distribution is:

$$P(S_{T+1} = j \mid \mathbf{y}_{1:T}) = \sum_{k=1}^K \hat{\alpha}_T(k) \cdot a_{kj} = [\hat{\boldsymbol{\alpha}}_T^\top \mathbf{A}]_j \quad (23)$$

For h -step-ahead: $\hat{\boldsymbol{\pi}}_{T+h|T} = \hat{\boldsymbol{\alpha}}_T^\top \mathbf{A}^h$.

5.2 Return Prediction

The predictive density is a mixture:

$$p(y_{T+1} \mid \mathbf{y}_{1:T}) = \sum_{k=1}^K P(S_{T+1} = k \mid \mathbf{y}_{1:T}) \cdot \mathcal{N}(y_{T+1} \mid \mu_k, \sigma_k^2) \quad (24)$$

Proposition 5.1 (Predictive Moments). *The predictive mean and variance are:*

$$\mathbb{E}[Y_{T+1} \mid \mathbf{y}_{1:T}] = \sum_{k=1}^K \hat{\pi}_{T+1|T}(k) \cdot \mu_k \quad (25)$$

$$\text{Var}(Y_{T+1} \mid \mathbf{y}_{1:T}) = \sum_{k=1}^K \hat{\pi}_{T+1|T}(k) (\sigma_k^2 + \mu_k^2) - \left(\sum_{k=1}^K \hat{\pi}_{T+1|T}(k) \mu_k \right)^2 \quad (26)$$

The variance decomposes into within-regime variance ($\sum \hat{\pi}_k \sigma_k^2$) and between-regime uncertainty ($\sum \hat{\pi}_k (\mu_k - \bar{\mu})^2$).

6 Empirical Application: Nifty 50

6.1 Data

Daily closing prices P_t of Nifty 50 index ($^{\wedge}\text{NSEI}$) from 2014-01-01 to 2026-04-10. Log-returns are computed as $Y_t = \log(P_t/P_{t-1})$.

6.2 Annualization

Reported statistics use the square-root-of-time rule with $\tau = 252$ trading days:

$$\mu_k^{\text{annual}} = \tau \cdot \hat{\mu}_k, \quad \sigma_k^{\text{annual}} = \sqrt{\tau} \cdot \hat{\sigma}_k \quad (27)$$

6.3 Regime Duration

The expected duration of regime k follows a geometric distribution:

$$\mathbb{E}[\text{duration} \mid S = k] = \frac{1}{1 - a_{kk}} \quad (28)$$

where a_{kk} is the probability of remaining in state k .

6.4 Occupancy Statistics

Empirical regime probabilities:

$$\hat{P}(S = k) = \frac{1}{T} \sum_{t=1}^T \mathbb{1}_{[s_t^* = k]} \quad (\text{hard}) \quad (29)$$

or

$$\hat{P}(S = k) = \frac{1}{T} \sum_{t=1}^T \gamma_t(k) \quad (\text{soft}) \quad (30)$$

7 Conclusion

The Hidden Markov Model provides a rigorous probabilistic framework for regime detection in financial time series. Key distinctions include:

1. **Smoothing** (Forward-Backward) for historical analysis using $\gamma_t(k)$
2. **Filtering** for real-time inference using $\hat{\alpha}_t(k)$
3. **Decoding** (Viterbi) for hard classification
4. **Prediction** using filtered probabilities and the transition matrix

The economic classification by mean return provides interpretable Bull/Bear/Crisis labels while maintaining statistical rigor.